Orientation Materials for the PQC Hard Problem Colloquium

Your Name

Your Institution / Research Group

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Purpose of the Colloquium

- This colloquium studies the computational assumptions underlying classical and post-quantum cryptography.
- Our focus is not merely on cryptosystems, but on the **hard problems** that sustain them.
- For each family, we ask three questions:
  1. What is the underlying hard problem?
  2. What are the best known attacks?
  3. How do classical and quantum complexities compare?

Why This Topic Matters

Central Observation

Cryptography survives only as long as its underlying hardness assumptions survive.

- Classical public-key cryptography relied mainly on factorization and discrete logarithms.
- Quantum algorithms, especially Shor's algorithm, fundamentally change this landscape.
- Post-quantum cryptography therefore asks which alternative problems remain hard even against quantum adversaries.

Classical Number-Theoretic Foundations

Historically, public-key cryptography was built on three dominant assumptions:

- **Integer Factorization** — RSA, Rabin
- **Discrete Logarithm in Finite Fields** — Diffie–Hellman, DSA, ElGamal
- **Elliptic Curve Discrete Logarithm** — ECDH, ECDSA, EdDSA

These problems were attractive because they supported efficient and elegant public-key constructions.

Integer Factorization

- Integer factorization underlies **RSA** and **Rabin**.
- Classically, the best known attacks include:
  - **Quadratic Sieve**,
  - **General Number Field Sieve**.
- These are **sub-exponential** classical attacks.
- In the quantum setting, **Shor's algorithm** solves the problem in polynomial time.

Orientation Point

Factorization is a model example of a problem that is hard enough classically, yet fundamentally broken quantumly.

Discrete Logarithm and Elliptic Curve DLP

Finite-field DLP

- Supports Diffie–Hellman, DSA, and ElGamal.
- Best classical attacks include **Index Calculus** and **Number Field Sieve**.
- These are again sub-exponential classically.

Elliptic-curve DLP

- Supports ECDH, ECDSA, and EdDSA.
- Generic classical attacks are roughly $O(\sqrt{n})$, e.g. Pollard's rho.
- Yet both finite-field DLP and ECDLP fall to **Shor's algorithm** in polynomial time.

The Transition to PQC

Key Transition

The post-quantum transition is the replacement of Shor-vulnerable assumptions with alternative assumptions for which no efficient quantum algorithms are known.

- This is not merely a standards transition.
- It is a transition in **mathematical foundations**.
- The main candidate families are:
  - lattices,
  - codes,
  - hash-based constructions,
  - multivariate systems,
  - isogenies.

Lattice Problems

The lattice family includes:

- **LWE**,
- **SIS**,
- **SVP**,
- **CVP**.

Representative systems include:

- Kyber / ML-KEM,
- Dilithium / ML-DSA,
- NTRU,
- fully homomorphic encryption constructions.

How Lattice Systems Are Attacked

The dominant attack language for lattice-based cryptography includes:

- **Lattice reduction**, especially LLL and BKZ,
- **Sieve algorithms**,
- careful **concrete security estimation**.

The best known attacks remain broadly **exponential** in the relevant dimension parameter.

Orientation Point

Lattice cryptography is currently the most important practical PQC family precisely because no Shor-style collapse is known.

Code-Based Problems

Code-based cryptography is centered on:

- **syndrome decoding**,
- more generally, decoding random linear codes.

Representative systems include:

- McEliece,
- HQC,
- BIKE.

The dominant attack family is **Information Set Decoding (ISD)** and its variants.

Code-Based Security Perspective

- Best known attacks remain **exponential**.
- Quantum improvements are comparatively limited.
- Grover's algorithm gives only a **quadratic speedup**, rather than a full structural break.

Orientation Point

Code-based cryptography remains one of the most conservative and durable post-quantum directions.

Hash-Based Assumptions

Hash-based cryptography relies on:

- collision resistance,
- preimage resistance,
- second-preimage resistance.

Representative signature systems include:

- SPHINCS+,
- XMSS,
- LMS.

Hash-Based Security Perspective

- Collision attacks follow the **birthday bound**.
- Quantum preimage search is accelerated by **Grover's algorithm**.
- But there is no analogue of Shor's algorithm causing polynomial-time collapse.

Orientation Point

Hash-based cryptography is often viewed as assumption-conservative, even when it is less attractive in size or efficiency.

Multivariate Problems

Multivariate cryptography is based on the difficulty of solving systems of quadratic equations over finite fields.

This is commonly framed as the **MQ problem**.

Representative systems include:

- UOV,
- MAYO,
- historically, Rainbow.

Multivariate Attack Perspective

Relevant attack language includes:

- **MinRank**,
- **Kipnis–Shamir**,
- **XL / Gröbner basis** methods,
- and scheme-specific structural attacks.

While the general problem remains hard-looking, particular constructions may fail because they reveal exploitable algebraic structure.

The Rainbow Lesson

- Rainbow was a prominent multivariate signature construction.
- It was broken by a polynomial-time structural attack.
- This teaches an important lesson:

Critical Distinction

Family-level hardness does not automatically imply scheme-level security.

Isogeny Problems

Isogeny-based cryptography is among the most mathematically elegant areas in PQC.

Representative names include:

- SQISign,
- CSIDH,
- historically, SIDH / SIKE.

The SIKE Lesson

- SIKE was broken by the **Castryck–Decru attack**.
- This attack was polynomial time and completely changed the status of SIDH/SIKE.
- Other attack frameworks include meet-in-the-middle methods and quantum algorithms such as Kuperberg's algorithm in certain commutative settings.

Orientation Point

Isogeny cryptography remains mathematically rich, but it must now be studied with exceptional caution regarding hidden structure.

Comparative Reading of the Landscape

The families divide naturally into four security narratives:

- **Factorization / DLP / ECDLP**: classically strong, quantumly broken.
- **Lattice / Code-based**: currently the strongest practical PQC foundations.
- **Hash-based**: conservative and structurally simple.
- **Multivariate / Isogeny**: mathematically compelling, but more exposed to structural fragility.

Three Questions for Every Scheme

For every construction discussed in this colloquium, we should ask:

1. What exact hard problem is assumed?
2. What are the best known classical and quantum attacks?
3. Does the scheme introduce exploitable structure beyond the underlying hard problem?

This provides a disciplined framework for comparing PQC proposals.

What the Hard-Problem Summary Really Teaches

The summary should not be read merely as a catalog of attacks.

It should be read as a set of research principles:

- distinguish **problem hardness** from **scheme security**,
- distinguish **generic attacks** from **structural attacks**,
- compare **classical** and **quantum** attack models,
- and study broken designs as lessons in cryptographic methodology.

Suggested Colloquium Flow

A natural progression for the colloquium is:

1. Classical number-theoretic assumptions and Shor's algorithm
2. Lattice problems and reduction-based security
3. Code-based assumptions and ISD
4. Hash-based signatures as a conservative baseline
5. Multivariate constructions and structural failure
6. Isogeny cryptography after SIKE
7. Cross-family comparison of security assumptions

Expected Outcomes

By the end of the colloquium, participants should be able to:

- explain the main hard-problem families in PQC,
- describe the best known attacks against each family,
- understand why some assumptions remain credible,
- and evaluate a cryptosystem at the level of its actual security assumption.

Closing Perspective

Final Perspective

Post-quantum cryptography is not simply the search for new algorithms. It is the search for mathematical problems whose hardness survives quantum computation, structural cryptanalysis, and concrete attack advances.

That is the central purpose of the PQC Hard Problem Colloquium.

A General Cryptographic Template

For each cryptographic family, we study a computational relation of the form

$$\mathcal{I} \mapsto \mathcal{S},$$

where:

- $\mathcal{I}$ is the public instance,
- $\mathcal{S}$ is the hidden witness, secret, or structure,
- and the adversary is asked to recover $\mathcal{S}$, or to solve a decisional variant.

Thus, a hard problem is typically one of the following:

$$\text{Search: } \mathcal{A}(\mathcal{I}) = \mathcal{S}, \qquad \text{Decision: } \mathcal{A}(\mathcal{I}) \in \{0, 1\},$$

with no efficient known algorithm for the relevant parameter regime.

Integer Factorization

Instance. Let

$$N = pq,$$

where p, q are large primes. The public input is N.

Search problem. Recover nontrivial factors:

$$\text{Given } N, \text{ find } p, q \text{ such that } N = pq.$$

Equivalent viewpoint. The hardness lies in inverting the map

$$(p, q) \longmapsto pq.$$

Why cryptographically useful. RSA relies on the asymmetry:

$$\text{multiplication is easy,} \qquad \text{factorization is hard.}$$

Attack language.

$$\text{QS,} \quad \text{GNFS,} \quad \text{Shor.}$$

Discrete Logarithm Problem in Finite Fields

Let $G = \langle g \rangle$ be a cyclic group, typically

$$G = \mathbb{F}_p^{\times}.$$

Instance. Given

$$g \in G, \qquad h = g^x \in G.$$

Search problem. Recover the exponent x:

$$\text{find } x \text{ such that } h = g^x.$$

Asymmetry. Exponentiation is easy:

$$x \mapsto g^x,$$

but inversion is believed hard.

Diffie–Hellman relation. Given

$$g^a,\ g^b,$$

compute

$$g^{ab}.$$

Elliptic Curve Discrete Logarithm Problem (ECDLP)

Let $E/\mathbb{F}_q$ be an elliptic curve, and let

$$P \in E(\mathbb{F}_q)$$

be a point of large order n.

Instance. Given

$$P, \qquad Q = xP.$$

Search problem. Recover

$$x \in \mathbb{Z}_n \quad \text{such that} \quad Q = xP.$$

Group law formulation. The map

$$x \longmapsto xP$$

is easy to evaluate, but hard to invert.

Generic attack baseline. Pollard's rho yields roughly

$$O(\sqrt{n}).$$

Exceptional structure. For special curves, one may reduce ECDLP to finite field DLP via

Lattice Problems: Geometric Form

A lattice generated by basis vectors $\boldsymbol{b}_1, \ldots, \boldsymbol{b}_m \in \mathbb{R}^n$ is

$$\Lambda = \left\{ \sum_{i=1}^{m} z_i \boldsymbol{b}_i : z_i \in \mathbb{Z} \right\}.$$

Shortest Vector Problem (SVP). Find

$$\boldsymbol{v} \in \Lambda \setminus \{\boldsymbol{0}\}$$

minimizing

$$\|\boldsymbol{v}\|.$$

Closest Vector Problem (CVP). Given $\boldsymbol{t} \in \mathbb{R}^n$, find

$$\boldsymbol{v} \in \Lambda$$

minimizing

$$\|\boldsymbol{t} - \boldsymbol{v}\|.$$

Interpretation. These are geometric optimization problems over a discrete subgroup of $\mathbb{R}^n$.

Learning With Errors (LWE)

LWE is an average-case algebraic-noise problem.

Secret.

$$\boldsymbol{s} \in \mathbb{Z}_q^n.$$

Samples. For random

$$\boldsymbol{a}_i \in \mathbb{Z}_q^n,$$

we observe

$$b_i = \langle \boldsymbol{a}_i, \boldsymbol{s} \rangle + e_i \pmod{q},$$

where e_i is a small error.

Search-LWE. Given many pairs

$$(\boldsymbol{a}_i, b_i),$$

recover $\boldsymbol{s}$.

Decision-LWE. Distinguish

$$(\boldsymbol{a}_i, \langle \boldsymbol{a}_i, \boldsymbol{s} \rangle + e_i)$$

from uniform samples in

SIS: Short Integer Solution

Let

$$A \in \mathbb{Z}_q^{m \times n}.$$

Problem. Find a nonzero vector

$$\boldsymbol{x} \in \mathbb{Z}^n$$

such that

$$A\boldsymbol{x} \equiv \mathbf{0} \pmod{q}$$

and

$$\|\boldsymbol{x}\|$$

is small.

Interpretation. SIS asks for a short relation among the columns of A modulo q.

Why important. SIS is a canonical source of hash functions and signature constructions.

Symbolic contrast with LWE.

LWE: hide $\boldsymbol{s}$ with noise, SIS: find short dependence modulo q.

Code-Based Hardness: Syndrome Decoding

Let

$$H \in \mathbb{F}_2^{r \times n}$$

be a parity-check matrix.

Given. A syndrome

$$\boldsymbol{s} \in \mathbb{F}_2^r$$

and a weight bound w.

Problem. Find an error vector

$$\boldsymbol{e} \in \mathbb{F}_2^n$$

such that

$$H\boldsymbol{e}^T = \boldsymbol{s}^T$$

and

$$\mathrm{wt}(\boldsymbol{e}) \leq w.$$

Interpretation. The vector $\boldsymbol{e}$ is sparse, but hidden by the linear map H.

Cryptographic asymmetry. Computing

Hash-Based Hardness

Let

$$H : \{0,1\}^* \to \{0,1\}^n$$

be a cryptographic hash function.

Preimage resistance. Given

$$y \in \{0,1\}^n,$$

find

$$x \text{ such that } H(x) = y.$$

Second-preimage resistance. Given x, find

$$x' \neq x \quad \text{such that} \quad H(x') = H(x).$$

Collision resistance. Find

$$x \neq x' \quad \text{such that} \quad H(x) = H(x').$$

Generic bounds.

collision search: $O(2^{n/2})$. preimage search: $O(2^n)$.

Multivariate Quadratic (MQ) Problem

Let

$$f_1, \ldots, f_m \in \mathbb{F}_q[x_1, \ldots, x_n]$$

be multivariate quadratic polynomials.

Problem. Solve the system

$$f_1(x_1, \ldots, x_n) = \cdots = f_m(x_1, \ldots, x_n) = 0.$$

Equivalently, for

$$\boldsymbol{x} = (x_1, \ldots, x_n),$$

find

$$\boldsymbol{x} \in \mathbb{F}_q^n$$

such that

$$F(\boldsymbol{x}) = \mathbf{0}.$$

Why difficult. Quadratic interactions create nonlinear constraints:

$$x_i x_j, \quad x_i^2.$$

Isogeny Problems

Let

$$E_1, E_2$$

be elliptic curves over a finite field.

Basic object. An isogeny is a nonconstant morphism

$$\phi : E_1 \to E_2$$

that is also a group homomorphism.

Hardness intuition. Given information about the endpoints, recover hidden isogeny data:

$$E_1 \rightsquigarrow E_2.$$

Depending on the system, the task may be:

- finding a path in an isogeny graph,
- recovering an ideal-class action,
- or reconstructing auxiliary hidden structure.

Symbolic viewpoint. The public data describes vertices in a graph,

A Unifying View

Each family can be summarized as "easy forward map, hard inverse problem":

$$
\begin{aligned}
(p, q) &\mapsto pq && \text{(Factorization)} \\
x &\mapsto g^x && \text{(DLP)} \\
x &\mapsto xP && \text{(ECDLP)} \\
(\boldsymbol{s}, \boldsymbol{e}) &\mapsto (A, A\boldsymbol{s} + \boldsymbol{e}) && \text{(LWE)} \\
\boldsymbol{x} &\mapsto A\boldsymbol{x} \bmod q && \text{(SIS / modular relations)} \\
\boldsymbol{e} &\mapsto H\boldsymbol{e}^T && \text{(Syndrome decoding)} \\
x &\mapsto H(x) && \text{(Hash-based)} \\
\boldsymbol{x} &\mapsto F(\boldsymbol{x}) && \text{(MQ)} \\
\phi &\mapsto \text{public isogeny data} && \text{(Isogenies)}
\end{aligned}
$$

The colloquium is therefore about one central theme:

Which inverse problems remain hard in the quantum era?

Thank you.

Questions and discussion